RESEARCH PROGRESS ADDENDUM / OCTOBER 17, 2026

Twelve new execution-backed runs. Better evidence, not yet task grades.

Adds ProgramBench **xsv** and **yj** experiments from commsfail PR #10 to the original nine-share study. The October 16 report and frozen evidence remain unchanged.

12
new Goal Run records

618 / 602
raw posts / agent posts

195
per-agent turn-log files

1,874
completed command events retained

Experiment design actually available

xsv: 2, 3, 4 agents; three repetitions each (9 runs).
yj: 2, 3, 4 agents; one repetition each (3 runs).
Episode-declared model: gpt-5.6-luna. Broadcast routing, isolated containers with shared workspace, consensus enabled, 30-minute / 3M-token stop limits. Runs occurred October 15, 2026.

Completion is not correctness

All 12 episodes record a `said DONE` stop. All runner check files are empty; hidden-test grades and final workspace snapshots were excluded upstream. **No success rate or quality gain can be established.**

TASK	AGENTS	RUNS	MEAN RAW POSTS	MEAN RUNNER MINUTES	MEAN RUNNER TOKEN COUNTER
xsv	2	3	27.0	6.11	143,485
xsv	3	3	52.7	3.45	186,739
xsv	4	3	67.7	4.83	255,404
yj	2	1	33.0	2.89	104,935
yj	3	1	66.0	6.21	210,850
yj	4	1	77.0	5.60	267,656

Descriptive means only, not a controlled estimate of the effect of agent count. Runner elapsed time is not summed agent work. Token counters are recorded metadata, not a billing calculation; cached counts are stored separately. Both cohorts together contain 21 traces across different tasks and evidence formats.

SAVED ANALYSES / SEPARATE PROGRAMBENCH COHORT

Every run analyzed; raw labels remain raw.

60 selected completed analyses: **36 substantive offline**, 12 demo-plugin examples, and **12 Inspect Scout** runs. Four facts_v1 rule flags are audited on the next page.

RUN / SOURCE ROOM	AGENTS	POSTS RAW/AGENT	RUNNER MIN	RUNNER TOKENS	FACTS FLAGS	DUP.	HANDOFF	INFO.
N1 xsv-n2-round1	2	23/22	13.73	226,106	0	Yes	Yes	Yes
N2 xsv-n2-round2	2	28/27	2.06	104,874	0	Yes	Yes	Yes
N3 xsv-n2-round3	2	30/29	2.54	99,475	0	Yes	Yes	Yes
N4 xsv-n3-round1	3	59/58	4.37	198,357	0	Yes	Yes	Yes
N5 xsv-n3-round2	3	43/42	2.89	172,680	0	Yes	Yes	Yes
N6 xsv-n3-round3	3	56/55	3.10	189,180	1	Yes	Yes	Yes
N7 xsv-n4-round1	4	61/60	3.29	269,373	0	Yes	Yes	Yes
N8 xsv-n4-round2	4	66/64	5.45	269,091	0	Yes	Yes	Yes
N9 xsv-n4-round3	4	76/74	5.73	227,748	0	Yes	Yes	Yes
N10 yj-n2-round1	2	33/32	2.89	104,935	1	Yes	Yes	Yes
N11 yj-n3-round1	3	66/64	6.21	210,850	0	Yes	Yes	Yes
N12 yj-n4-round1	4	77/75	5.60	267,656	2	Yes	Yes	Yes

These Yes/No cells are model judgments, not verified failures.

duplicated work: 12/12, weak handoff: 12/12, unintegrated information: 12/12. The categories are duplicated work, weak handoff, and unintegrated information. A deterministic citation audit also finds **20/36 records** whose narrative [M#] citations are not fully represented by their structured original-turn list. Raw outputs are preserved, not silently corrected; use original room evidence, not unverified click-through citations, for conclusions.

Actual saved versions: 7 runs: 2.1.0 / prompt 3; 4 runs: 2.1.1 / prompt 4; 1 runs: 2.1.4 / prompt 7; model `openai/gpt-5.4-mini`. Earlier successes were reused rather than rerun solely to homogenize versions. 8 failed Scout attempts are retained in history, separate from the 60 selected completions. Demo output is not another validated detector.

QUALITATIVE EVIDENCE AUDIT / NOT A GOLD-LABEL STUDY

Logs make the detector's mistakes inspectable.

facts_v1 emits four flags across three runs. Reviewing the cited evidence does **not** establish four collaboration failures: three contradiction flags are unsupported by the cited text; one status flag is ambiguous shell-control behavior.

RAW FLAG / ORIGINAL MESSAGE	WHAT THE RETAINED EVIDENCE SUPPORTS
N6 #48 `success_after_failure`	The cited command is a smoke-test reporting loop ending in a conditional echo. Its exit status can be 1 when the last condition is false. Its full output is empty; post #47 reports no crashes. This is not evidence of a hidden-test failure.
N10 #31 `private_contradiction`	The cited same-turn text announces a plan to check error behavior, not a failed result. The actual final harness message is DONE. The broad error-keyword rule mistakes a test plan for contradiction.
N12 #68, #69 `private_contradiction`	The cited harness text describes expected exit-code-1 validation for an invalid -k configuration, alongside successful builds. Mentioning the expected error is not a contradiction of completion. These are two rule flags, not two verified failures.

Independent checking can prevent false closure

In yj / 4 agents, posts #62–64 say DONE. Another agent then reports the invalid –k behavior at #65, reports a fix at #66, and the team rechecks it at #67–72. Repeated DONE and verification can follow the consensus protocol, not necessarily indicate wasted work.

Original posts #62–72. These are conversational reports, not independent hidden-test grades.

Scout needs the same scrutiny

The yj / 4-agent duplicated-work explanation cites repeated validation and checks after fixes. Its definition excludes deliberate independent verification. Treat this positive as review material, not proof of unnecessary duplication.

This is our source-based reading, not blinded independent annotation, adjudicated labels, or a precision estimate. Zero facts_v1 flags in nine other runs is not evidence of success.

METHOD / COVERAGE / LIMITATIONS

From immutable inputs to inspectable results.

1. Pin and verify inputs

PR #10 head `913ad1805642a38e5248d67e5d8c3b83c48b51b5`. Verify 255 raw files against the two original byte-length / SHA-256 manifests. Preserve source bytes and task instructions; do not execute commands from logs.

2. Normalize without silently losing evidence

Split JSONL on newline, preserving U+2028 inside strings. Link every message ID returned by compound say commands. Regression tests recover one completed command and five DONE links: **602/602 agent posts linked; 1,874/1,874 command events retained.**

3. Run complementary analyses

Structural heuristics and regex are offline signals. facts_v1 compares messages with per-seat operations. Scout annotates communicative function, selects candidate context, then judges three collaboration categories from room text—not execution logs.

4. Preserve successes and explicit errors

Save records, prompts, annotations, versions, timestamps, and 8 failed attempts. Improve rejected-span feedback while retaining exact-substring validation and the three-attempt bound. Native Scout replay uses saved judgments without new inference.

New Scout coverage: 602 annotations; 602 candidate messages; 0 count-truncated traces. 0 agent messages exceed 2,000 characters; 0 exceed 8,000. Candidate text is capped at 2,000 characters each; annotation targets at 8,000, with up to six cited prior messages × 2,000 and twelve preceding messages × 500. A false count-truncation flag does not imply full-text visibility.

Important comparability limits: Goal/runner messages are excluded here; earlier share-only Scout outputs included task posts because roles were unknown. [M#] display labels differ from original turn IDs; saved native replay preserves structured IDs and does not repair the 20 inconsistent narrative records. facts_v1 uses filename matching, coarse compound-command boundaries, and some whole-run/same-turn evidence; it is not a causal or strictly online predictor. Normalized command output retains its last 500 characters; full streams remain. Full sessions, workspace histories, task grades, and Scout generation/token event logs are unavailable.

Next controlled study: obtain evaluator revision and submission hashes/grades; repeat matched tasks and budgets across agent counts; freeze reader/analyzer/prompt versions; label both positives and negatives with independent raters, explicitly distinguishing verification from duplication. No training improvement has been measured here.

Use PRs for review. Give each run one durable home.

REPOSITORY	PRIMARY ROLE	WHAT BELONGS THERE
multi-agent-training	Research coordination	Proposal, protocols, conditions, evaluation plan, experiment catalog and analysis links. Team access is required.
commsfail	Reusable analysis engine	Trace readers, annotators, schemas/taxonomies, audit tooling, and explicitly approved representative fixtures.
agenttrace	Application, inspection, reporting	Web UI/API, SQLite history, batch exports, Inspect Scout integration, saved-result replay, approved cohort snapshots and reports.

AgentTrace imports commsfail under `packages/commsfail`; upstream changes are not automatically synchronized. Record the actual analysis code revision.

A practical workflow

Raw run: immutable bundle, stable run ID, source revision and SHA-256.
Research catalog: small reviewed manifest linking conditions, evaluator and artifacts.
Analysis release: versioned outputs, annotations, errors and report; SQLite is a convenient frozen viewing export.

Keep small approved text collections in Git if practical. Move growing bulk archives to a designated dataset release or artifact store, with checksummed references rather than repeated copies in all three repos.

PR #10 is useful, with a caveat

It makes a small public cohort reviewable and hash-verifiable. But a PR is not an archival policy, and an unmerged PR is not upstream main. The sample README permits real samples while CONTRIBUTING still says experiment data stays outside the library; clarify that exception.

This publication is a checksummed mirror, not twelve additional experiments. No hosting or new storage service was created.

Capture next time: task/runner/container revisions, resolved models, seed, roles/routing/budgets, stop reason, complete logs, approved final workspace/patch history, evaluator revision and grade, and hashes of submitted artifacts. Never publish agent homes, credentials, capability links, or protected hidden tests.

Detailed storage guidance and example manifest · Pinned upstream contribution/data policy · GitHub release documentation

Open the same 21 traces without new model calls.

NEW RUN	CANONICAL TRACE ID	SELECTED SCOUT RUN ID
N1 xsv-n2-round1	`trace_810039cd98b3c8b0`	`run_1bd4978afe184d78a0eaa7aa6a5e295b`
N2 xsv-n2-round2	`trace_f18bef1f6956af75`	`run_9b9c8d11c359411c8914209075f73875`
N3 xsv-n2-round3	`trace_3ce5b38eb14b6137`	`run_3b23f3c16acc4cfcb234d059a82831bc`
N4 xsv-n3-round1	`trace_f1509e7ad764d54a`	`run_e07c60b29c80403095d7602f3a4bff68`
N5 xsv-n3-round2	`trace_abb6da728982d695`	`run_899cc6b3861d4e039bea31060609a964`
N6 xsv-n3-round3	`trace_b7ff2953e20e3e5e`	`run_2a7cb52622df49659381ec3800095eea`
N7 xsv-n4-round1	`trace_a49b03f1df3bca94`	`run_e713662b225f4f6c9d6f2110bb36d2b7`
N8 xsv-n4-round2	`trace_abaad9e890b53c8a`	`run_37c0dd9f10984622a8c19d7ba1aaf4fb`
N9 xsv-n4-round3	`trace_f4edfa23b99faa29`	`run_25f49ea33b77439f99448b3ae36b693c`
N10 yj-n2-round1	`trace_b128ff4c92d47fa5`	`run_3778b64258a64d2888c05a18658b0d0f`
N11 yj-n3-round1	`trace_c25e8be963216dac`	`run_418e89a41e664bd38d6b034e93ae8f96`
N12 yj-n4-round1	`trace_ae5cc6010f6d3028`	`run_d84ba8d5d7fc4a3f99438897ea3ebeb0`

```
python3 tools/restore_results.py data/teammate-traces data/programbench-pr10 \
  --output .agenttrace/all-teammates
AGENTTRACE_DATA_DIR="$PWD/.agenttrace/all-teammates" \
  AGENTTRACE_SCOUT_VIEWER_URL=http://127.0.0.1:7577 PORT=3001 agenttrace
```

After installing dependencies and building the frontend, run from the repository root. Restore refuses to overwrite an existing working copy. The new cohort snapshot contains 68 historical runs; 60 are selected completions. The combined viewer includes 96 selected results across 21 traces. Open `data/programbench-pr10/batch/all-cohorts.html` locally for all result links. Start native Scout on port 7577 as documented and click **Open in Scout**; do not click Run Scout merely to view saved results.

Published evidence

Original PR · New cohort · Summary / all selected run IDs · Checksums

Original study, preserved

October 16 professor report · Nine-share snapshot

This HTML is self-contained; download and open locally. GitHub's file view is not a hosted application.